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ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech (EE)

Semester: V

Name of the Paper: Microprocessors

Paper Code: **TEE 503**

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Draw the Pin Diagram of 8085 Microprocessor and discuss each pin in detail.	
(b)	Explain address bus, data bus and control bus of 8085 microprocessor. Draw system bus architecture of 8085 microprocessor.	
(c)	Draw and define the Architecture of 8085 Microprocessor.	
Q2	(20 marks)	CO2
(a)	Write the difference between Memory-mapped I/O and Peripheral I/O.	
(b)	Explain the addressing modes of 8085 microprocessor with examples.	
(c)	Write an assembly language program in 8085 to perform multiplication of two 8-bit numbers.	
Q3	(20 marks)	CO3
(a)	Explain 8237 DMA controller along with diagram.	
(b)	Explain 8251 Programmable Communication Interface.	
(c)	Discuss 8254 programmable Timer along with diagram.	
Q4	(20 marks)	CO4
(a)	Write an assembly language program in 8086 to perform division of 32-bit number by 16-bit number.	
(b)	Draw and define the Architecture of 8086 Microprocessor.	
(c)	Write an assembly language program to perform sum of two 8-bit BCD number with carry.	
Q5	(20 marks)	CO5
(a)	Explain the addressing modes of 8086 microprocessor with examples.	
(b)	Explain shift and rotate instructions of 8086 microprocessor.	
(c)	Explain ADC and DAC in detail.	

